

Emre Kerman

Computer Engineer

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SUMMARY

I'm a computer engineer from İzmir, freshly graduated from Dokuz Eylül University, who loves linux and open-source software. Experienced in software development, machine learning, and embedded systems, with hands-on experience in IoT, computer vision, web technologies, both frontend and backend development. Always eager to learn new technologies and contributing.

EDUCATION

Dokuz Eylül University

Bachelor of Science in Computer Engineering

İzmir, Turkey

2021 – 2025

Yunus Emre Anadolu Lisesi

High School Diploma

İzmir, Turkey

2018 – 2021

EXPERIENCE

Software & IoT Engineering Intern

İndas Teknoloji Ticaret AŞ.

June 2025 – Sept. 2025

İzmir, Turkey

- Engineered a smart parking platform with ESP32-CAM nodes and OCR (PaddleOCR/Tesseract) for automated entry tracking, billing, and balance management via Next.js REST APIs.
- Built an edge-first PPE and human detection system running quantized FOMO models on ESP32-CAM (120–250ms latency) using Edge Impulse and C++/Arduino.
- Designed dual-buffer architecture and PSRAM optimizations on ESP32-CAM to achieve 18 FPS live streaming and 5–6 FPS on-device inference.
- Developed web frontends using Next.js, React, and Tailwind CSS for real-time bounding box overlays, detection logs, and balance dashboards.

Software Engineering Intern

Dokuz Eylül University Computer Engineering Dept.

Nov. 2025 – Dec. 2025

İzmir, Turkey

- Developed "Restock", a phygital self-checkout kiosk platform syncing physical store kiosks with mobile devices for real-time session handover.
- Implemented atomic DB transactions using Drizzle ORM and SQLite alongside custom React hooks for hardware/browser barcode scanning.
- Configured role-based auth (JWT) via Next.js Middleware and built administrative monitoring dashboards achieving 90+ Lighthouse scores.

PROJECTS

Early Frost Detection | Go, Python, C, TensorFlow, Keras, NumPy, ESP-32

2024 – 2025

- Developed an AI-driven IoT solution to predict ice formation on refrigerator cooling fins for Arçelik.
- Built a real-time web server and front-end monitoring platform using Go.
- Designed and assembled a hardware refrigerator prototype for accurate experimental data collection.
- Configured ESP-32 microcontrollers and DHT-22 sensors to collect environmental data over Wi-Fi.
- Performed data preprocessing, feature engineering, and predictive model training.

Hand Gesture Detection | Python, TensorFlow, Keras, NumPy, OpenCV

2024

- Constructed two computer vision models (a lightweight CNN and a ResNet-like architecture) to classify gestures via webcam.
- Curated and preprocessed a custom dataset from scratch.
- Implemented an efficient training pipeline utilizing Model Checkpoints to prevent data loss.
- Integrated live webcam input with UX enhancements for real-time inference.

Game Platform & Storefront | C#, .NET Core, PostgreSQL

2023

- Designed and implemented the complete database architecture for a digital game distribution platform.
- Developed backend application logic to support user authentication, game publishing, and purchasing.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, Go, Python, C/C++, C#, Java, SQL, Lua

Web & Frameworks: Next.js, React, React Native, Astro, .NET Core, Tailwind CSS, shadcn/ui, REST APIs

Databases & Backend: PostgreSQL, SQLite, Drizzle ORM, Auth (OAuth/JWT), Swagger / OpenAPI

AI, Vision & Edge IoT: TensorFlow, Keras, TFLite, OpenCV, Edge Impulse, ESP32 / ESP32-CAM, PaddleOCR

DevOps & Systems: Linux, Nix / NixOS, Docker, Kubernetes, Git, Intel Quartus Prime

Languages Spoken: Turkish (Native), English (Advanced - C1), German (Beginner - A1)